Aditya Kumar Raj

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CAREER OBJECTIVE

Passionate and highly motivated B.Tech Computer Science (Artificial Intelligence) undergraduate with hands-on experience in Python, Machine Learning, Computer Vision, Generative AI, IoT, and Robotics. Skilled in developing AI-powered solutions using deep learning frameworks, edge AI systems, and real-time data processing. Seeking an entry-level AI/ML Engineer position to contribute to intelligent and scalable solutions while continuously exploring emerging AI technologies.

EDUCATION

Bachelor of Technology (B. Tech) – Computer Science & Engineering (Artificial Intelligence)
Babu Banarasi Das University, Lucknow | 2022 - 2026
Last Semester SGPA: 8.27

Intermediate (Class XII) – Science
Uttar Pradesh Madhyamik Shiksha Parishad | 2022

High School (Class X)
Central Board of Secondary Education | 2020

TECHNICAL SKILLS

Programming Languages

- Python

Artificial Intelligence & Machine Learning

- Machine Learning
- Computer Vision
- Natural Language Processing (NLP)
- Generative AI & Prompt Engineering
- Deep Learning Fundamentals

Libraries & Tools

- NumPy
- Pandas
- Data Analysis
- Data Visualization
- Streamlit
- Excel

Web Technologies (Front-End Development)

- HTML
- CSS
- JavaScript

PROJECTS

AI & IoT-Powered Plant Disease Detection and Removal System Using Computer Vision Technologies | Feb 2026 – May 2026

- Developed an AI-powered precision agriculture system for real-time plant disease and environmental threat detection.
- Built hybrid cloud-to-edge architecture integrating YOLOv8 and FOMO-based Edge AI models.
- Trained models on 191,000+ images spanning 84 classes using automated annotation and optimization pipelines.
- Achieved approximately 89.2% mAP for object detection tasks.
- Integrated ESP32, ESP XIAO, DHT11, soil moisture sensors, OLED display, and motor control systems.
- Developed Streamlit dashboards for live telemetry, AI inference, and treatment recommendations.
- Reduced deployment cost by nearly 75% through low-cost edge AI architecture.

Laser-Based Vertical Height Measuring Robot | Mar 2024 – Apr 2024

- Designed and developed an Arduino-based robotic system utilizing laser sensors and MPU-6050 IMU.
- Implemented tilt-compensation algorithms improving measurement accuracy by 80–85%.
- Performed real-time sensor data processing for stable height estimation.
- Applied Robotics, Embedded Systems, IoT, and Sensor Fusion concepts to solve real-world engineering problems.

TRAINING

- **Core Java & Advanced Java Training Program**
  Grastech Pvt. Ltd., Lucknow | *Jun 2025 – Jul 2025*

- **Soft Skill Training Program**
  Learnovate Enterprises, Lucknow | *Mar 2025 – Apr 2025*

CERTIFICATIONS

- **Certificate of Participation – TechXplore**
- **Certification of completion course in - Artificial Intelligence**
- **Certification of completion course in - Python Programming**

SOFT SKILLS

- **Problem Solving**
- **Analytical Thinking**
- **Team Collaboration**
- **Communication Skills**
- **Time Management**
- **Adaptability**
- **Decision Making**

LANGUAGES

- **English**
- **Hindi**

AVAILABILITY

Open to Internships and Full-Time Opportunities in:

- **AI Engineer**
- **Machine Learning Engineer**
- **Computer Vision Engineer**
- **Data Science Intern**
- **Generative AI Engineer**
- **Software Engineer**